

KINGSTON 1 W, RI, USA

WMO: 725120

Lat: 41.4782N

Lon: 71.5417W

Elev: 32

StdP: 100.94

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54797

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.3	-22.4	0.5	-13.0	-19.6	0.7	-10.4	8.6	-0.1	7.6	-0.4	1.7	N/A	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.1	23.2	28.5	22.3	27.1	21.4	24.6	28.4	23.7	26.9	23.0	25.8	2.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.3	26.6	22.7	17.5	25.6	22.0	16.7	24.7	74.2	28.5	70.8	27.0	68.2	25.7	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.1	5.3	DB	-19.4	32.9	3.6	1.5	-22.0	34.0	-24.1	34.9	-26.1	35.7	-28.8	36.8	(4)
			WB	-20.5	26.4	3.4	1.3	-22.9	27.3	-24.9	28.1	-26.8	28.8	-29.2	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.0	-1.2	2.5	8.1	13.2	17.9	21.6	20.8	17.2	11.2	5.8	1.3
	DBStd	9.19	5.69	4.99	4.69	3.88	3.75	3.34	2.63	2.89	3.51	4.47	4.62	4.98
	HDD10.0	1469	373	313	237	82	10	0	0	0	0	39	142	273
	HDD18.3	3392	632	546	492	307	164	47	4	9	62	223	377	529
	CDD10.0	1386	1	1	3	26	111	237	361	336	217	77	16	3
	CDD18.3	266	0	0	0	1	6	34	107	87	28	3	0	0
	CDH23.3	1823	0	0	0	14	58	265	737	605	133	12	0	0
	CDH26.7	397	0	0	0	3	11	59	178	129	16	1	0	0
Wind	WSAvg	2.2	2.6	2.6	2.8	2.6	2.1	1.7	1.6	1.5	1.7	2.1	2.2	2.4
Precipitation	PrecAvg	1300	110	97	125	118	107	93	79	105	109	105	127	121
	PrecMax	1783	315	211	312	329	244	364	187	323	276	379	330	286
	PrecMin	824	23	23	18	31	18	1	19	14	23	16	19	23
	PrecStd	200	58	43	62	62	55	73	47	68	54	62	64	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	13.4	18.7	25.1	28.2	31.1	32.4	31.8	28.7	25.3	19.0	15.8
		MCWB	12.4	10.2	12.6	17.5	21.2	23.4	24.6	24.1	23.5	18.9	15.6	13.9
	2%	DB	11.5	11.0	14.0	20.2	24.8	28.3	30.3	29.6	26.3	21.9	17.2	13.8
		MCWB	9.9	8.7	9.8	13.7	18.8	21.3	23.5	22.8	21.1	19.3	15.0	12.6
	5%	DB	9.0	9.0	11.8	17.2	22.4	26.3	28.7	28.0	24.8	20.2	15.7	12.0
		MCWB	7.5	6.8	8.9	12.2	17.2	20.2	22.6	21.9	20.5	17.6	13.2	10.8
	10%	DB	6.6	6.7	9.9	15.0	20.1	24.5	27.2	26.6	23.5	18.8	14.2	10.0
		MCWB	4.7	4.4	7.1	10.9	15.6	18.8	21.9	21.5	20.1	16.4	11.7	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.5	15.0	17.0	22.2	24.1	26.3	25.6	24.2	21.1	17.0	15.1
		MCDB	13.9	13.4	18.7	21.4	27.2	29.0	31.4	29.5	28.0	22.8	18.3	15.5
	2%	WB	10.4	10.1	11.1	14.8	20.4	22.6	24.9	24.2	23.0	19.8	15.7	13.3
		MCDB	11.1	11.3	13.0	16.7	23.9	26.9	29.3	27.2	25.4	21.5	16.6	14.0
	5%	WB	7.7	7.8	9.7	13.4	18.7	21.4	23.9	23.5	21.9	18.4	13.9	11.6
		MCDB	9.0	9.3	11.8	15.7	21.6	24.7	27.5	26.0	23.7	20.3	15.2	12.4
	10%	WB	5.2	5.4	7.4	11.8	17.3	20.3	23.2	22.8	21.1	17.2	11.6	9.3
		MCDB	6.5	7.4	9.4	13.6	19.8	23.1	26.3	25.0	22.8	18.8	13.5	10.7
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.7	9.7	10.5	10.3	10.3	9.9	10.4	10.7	10.5	10.3	9.1
		MCDBR	11.6	11.8	12.1	14.1	13.3	12.8	11.6	11.4	11.4	10.9	10.9	10.3
		MCWBR	11.3	9.5	8.0	7.3	7.4	7.0	5.7	5.9	6.8	7.6	9.7	10.1
	5% WB	MCDBR	12.0	11.9	10.5	9.6	11.0	10.8	9.9	9.2	8.4	9.1	10.3	10.1
		MCWBR	11.7	10.8	8.4	7.1	7.0	6.4	5.3	5.5	5.8	7.2	10.0	10.3
Clear-Sky Solar Irradiance	taub	0.305	0.313	0.336	0.374	0.419	0.461	0.486	0.445	0.376	0.344	0.315	0.303	
	taud	2.500	2.477	2.453	2.381	2.288	2.208	2.149	2.254	2.421	2.519	2.557	2.556	
	Ebn at Noon	852	900	914	897	860	821	795	818	859	853	836	828	
	Edh at Noon	74	87	100	116	131	142	149	130	103	83	70	65	
All-Sky Solar Radiation	RadAvg	1.89	2.76	3.78	4.88	5.40	5.92	6.06	5.44	4.44	3.00	2.05	1.58	
	RadStd	0.15	0.27	0.27	0.52	0.52	0.59	0.32	0.29	0.33	0.28	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	+47

Nomenclature: See separate page